Dr. Shivam Singh (Ph.D.)

Postdoctoral Research Associate, University of Virginia, VA, USA

Department of Civil Engineering, Indian Institute of Technology Indore

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Career goal

My career goal is to contribute to next-generation climate science by improving how extreme hydro-meteorological events are understood, modeled, and translated into actionable information for society. I aim to build research that bridges climate dynamics, high-resolution environmental modeling, and modern artificial intelligence, with an emphasis on developing reliable methods that generalize across regions, climate regimes, and evolving risk landscapes. In the long term, I aspire to become an independent researcher and educator, leading interdisciplinary collaborations and mentoring students to address urgent challenges in climate risk and resilience.

Academic records

Ph.D.	**Civil Engineering** — **Indian Institute of Technology Indore, M.P., India** July 2019 - November 2023
M.Tech 8.03/10	**Civil Engineering** — **Maulana Azad National Institute of Technology, M.P., India** July 2017 - June 2019
B.Tech 7.82/10	**Civil Engineering** — **Maulana Azad National Institute of Technology, M.P., India** July 2013 - June 2017
12th class 86.00%	**Science** — **Chitrakoot Inter College, Chitrakoot, Uttar Pradesh, India** June, 2011
10th class 79.00%	**Science** — **G.D.N.D.S.N. Inter College, Chitrakoot, Uttar Pradesh, India** June, 2009

Experiences

August 01, 2024 -Present	Advanced AI for Simultaneous Bias-Correction and Down-scaling of Climate Model Projections *Postdoctoral Research Associate* Environmental Institute, University of Virginia, USA
July 24, 2023 -July 14, 2024	Technological Innovation and Intellectual Property *Project Associate I*, Department of Civil Engineering, IIT Indore

Articles

1. **Singh, S.**, Goyal, M. K. & Jha S. (2023). Role of large-scale climate oscillations in precipitation extremes associated with atmospheric rivers: nonstationary framework. *Hydrological Sciences Journal*, 68(3), 395-411.
2. **Singh, S.** & Goyal, M. K. (2023). Enhancing climate resilience in businesses: The role of artificial intelligence. *Journal of Cleaner Production*, 418, 138228.
3. **Singh, S.** & Goyal, M. K. (2023). An innovative approach to predict atmospheric rivers: Exploring convolutional autoencoder. *Atmospheric Research*, 289, 106754.
4. **Singh, S.**, Yadav A. & Goyal, M. K. (2024).Univariate and bivariate spatiotemporal characteristics of heat waves, and relative influence of large-scale climate oscillations over India. *Journal of Hydrology*, 628, 130596.
5. **Singh, S.**, Kumar N., Goyal, M. K. & Jha S. (2023). Relative influence of ENSO, IOD, and AMO over spatiotemporal variability of hydroclimatic extremes in Narmada basin, India. *AQUA - Water Infrastructure Ecosystems and Society*, 72(4), 520-539.
6. **Singh, S.**, Goyal, M. K. (2023), & Saikumar E. Assessing Climate Vulnerability of Ramsar Wetlands through CMIP6 Projections. *Water Resources Management*, (2024).
7. **Singh, S.**, Prasad B., & Tiwari, H.L. (2021). Sedimentation analysis for a reservoir using remote sensing and GIS techniques. *ISH Journal of Hydraulic Engineering*, 29(1), 71-79.
8. Meghani, S. **Singh, S.**, Kumar, N. & Goyal, M. K. (2023). Predicting the spatiotemporal characteristics of atmospheric rivers: A novel data-driven approach. *Global and Planetary Change*, 231, 104295.
9. Kumar, N., Patel, P., **Singh, S.** & Goyal, M. K. (2023). Understanding non-stationarity of hydroclimatic extremes and resilience in Peninsular catchments, India. *Scientific Reports (Nature)*, 13, 12524.
10. Gupta V., Rakkasagi S., Rajpoot S., El Immanni H. S. & **Singh S.** (2023). Spatiotemporal analysis of Imja Lake to estimate the downstream flood hazard using the SHIVEK approach. *Acta Geophysica*, 71, 2233-2244.

11. Goyal, M. K., **Singh, S.** & Jain, V. (2023). Heat waves characteristics intensification across Indian smart cities. *Scientific Reports (Nature)*, 13, 14786.
12. Rautela, K. S., **Singh, S.** & Goyal, M. K. (2024). Characterizing the spatiotemporal distribution, detection, and prediction of aerosol atmospheric rivers on a global scale. *Journal of Environmental Management*, 315, 119675.
13. Rautela, K. S., **Singh, S.** & Goyal, M. K. (2024). Aerosol atmospheric rivers: patterns, impacts, and societal insights. *Environmental Science and Pollution Research*, 315, 119675.
14. Rautela, K. S., **Singh, S.**, & Goyal, M. K. (2024). Resilience to Air Pollution: A Novel Approach for Detecting and Predicting Aerosol Atmospheric Rivers within Earth System Boundaries. *Earth Systems and Environment*, 9, 93–115.
15. Abdelmoaty, M. H., Papalexiou, S. M., Mamalakis, A., **Singh, S.**, Coia, V., Hairabedian, M., Szeftel, P., & Grover, P. (2025). Generative Adversarial Networks for Downscaling Hourly Precipitation in the Canadian Prairies., *Journal of Geophysical Research: Machine Learning and Computation*, 2, e2025JH000678.
16. **Singh, S.**, Abdelmoaty, M. H., Papalexiou, S. M., Hartvigsen, T., & Mamalakis, A. (2025). Correcting Dry/Wet Classification Bias in Precipitation Downscaling via Generative Adversarial Networks. *Environmental Data Science* (Under review).
17. Abdelmoaty, M. H., Papalexiou, S. M., Mamalakis, A., **Singh, S.**, Coia, V., Hairabedian, M., Szeftel, P., & Grover, P. (2025). Does Non-Stationarity Affect GAN-Based Downscaling? Insights from High-Resolution WRF Simulations over the Canadian Prairies, *Artificial Intelligence for the Earth Systems* (Under review).
18. **Singh, S.**, Abdelmoaty, M. H., Papalexiou, S. M., Hartvigsen, T., & Mamalakis, A. (2026). Comprehensive Intercomparison of Generative AI Models for Super-Resolution Precipitation Downscaling Across Hydroclimatic Regimes. *Water Resources Research* (Under review).

Patent(s)

1. Goyal, M. K. & **Singh, S.**. (2023). Method and system for determining spatiotemporal characteristics and predicting the state of Atmospheric Rivers (AR). Filed in the Indian Patent Office on February 15, 2024. Application No. 202421010646.

Book(s)

1. Goyal, M. K. & **Singh, S.**. Understanding Atmospheric Rivers Using Machine Learning. *SpringerBriefs in Applied Sciences and Technology* ISBN: 978-3-031-63477-2. Springer.

Book chapters

1. Vivekanandan, N., **Singh, S.**, & Goyal, M.K. (2023). Comparison of Probability Distributions for Extreme Value Analysis and Predicting Monthly Rainfall Pattern Using Bayesian Regularized ANN. In *Ecosystem Restoration: Towards Sustainability and Resilient Development*. (pp. 271-294). Springer.
2. **Singh, S.**, Dubey, S., Kumar, N., Goyal, M. K., & Pal, I. (2020). Psychological Impacts of COVID-19. Integrated Risk of Pandemic: Covid-19 Impacts, Resilience and Recommendations, 153–168.

Conferences

1. American Geophysical Union 2025, New Orleans, USA.
   Singh, S., Abdelmoaty, M. H., Papalexiou, S. M., Hartvigsen, T., & Mamalakis, A. Generative Models for Super-Resolution Precipitation Downscaling Across U.S. Hydroclimatic Regimes. Poster Presentation, December 15-19, 2025.
2. National Symposium on Tropical Meteorology, India.
   Singh, S. and Goyal M. K. *Heat waves in India: Past and Future perspectives* Indian Meteorological Society and IISER Bhopal, Madhya Pradesh. Oral Presentation, November 29 -December 2, 2022.
3. International Conference on Smart Systems and Advanced Computing (SysCom 2022), China.
   Singh, S. and Goyal, M. K. *Predicting Atmospheric Rivers Employing a Deep Learning Algorithm: Autoencoder* Macau, China, Oral Presentation (Virtual Mode), December 30-31, 2022.
4. 36th Madhya Pradesh Young Scientist Congress, India.
   Singh, S. and Goyal, M. K. *Role of large-scale climate oscillations over hydroclimatic extremes of Narmada Basin* Madhya Pradesh Council of Science and Technology. Oral Presentation, March 23-26, 2021.
5. 26th International Conference on Hydraulics, Water Resources, Environmental and Coastal Engineering (HYDRO 2021), India.
   Singh, S. and Goyal, M. K. *Heat waves in India: Past and Future perspectives* National Institute of Technology Rourkela. Oral Presentation, March, 2021.

Achievements/ Awards

- **Young Scientist Award** by Madhya Pradesh Council of Science and Technology, Bhopal in the 36th Madhya Pradesh Youth Science Congress (2021).
- **Best Research Synopsis Award** in PhD Colloquium 2022 jointly organized by Atal Bihari Institute for Good Governance and Policy Analysis (AIGGPA), Bhopal and Indian Council of Social Science Research (ICSSR), New Delhi.
- **Best Poster Presentation Award** in Research and Industrial Conclave (2022) organized by Indian Institute of Technology Indore.
- **Second Place in the 2025 Research Showcase Competition** during the 2025 Postdoctoral Research Symposium: "Bridging Gaps: Fostering Belonging in the Postdoc Community", held on May 21, 2025, at the University of Virginia Alumni Hall, Charlottesville, Virginia, USA.
- **Prime Minister Scholarship** for meritorious wards of Central Armed Forces awarded by the Central Government of India.
- **Secured 14045th rank** in all India in Joint Entrance Exam **(JEE) Main 2013**, out of **12.8 lakh** registered students.

Research interests

- Hydro-meteorological Extremes
- Climate Variability and Change
- Atmospheric Rivers
- Climate Downscaling and Bias Correction
- Machine Learning for Earth System Science

Responsibilities

- Vice-Chair of UVA Postdoc Advocacy Committee
- Teaching Assistant, IIT Indore, 2019-23
- Coordinated Training Programs, IIT Indore, 2022-23
- INSPIRE Club head, NIT Bhopal, 2016-17
- Volunteer as Judge for Departmental Symposiums

Proficiency

Technical Expertise

- Climate Data Handling
- Hydrological Modeling
- Machine Learning

Analytical Techniques

- Statistical Analysis
- Probabilistic Analysis
- Risk Mapping

Software Packages

- R Studio, Python, MATLAB
- Microsoft Office
- ArcGIS, SWAT

Languages

- Hindi ★★★★★
- English ★★★★★
- Sanskrit ★★☆☆☆

Hobbies

- Reading Ancient Vedic Texts
- Learning New Software
- Memorizing Sanskrit Verses

References*

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Dr. Antonios Mamalakis
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Hereby, I declare that all the information furnished above is true.

Date : January 28, 2026
Place : Charlottesville, Virginia

Shivam Singh